

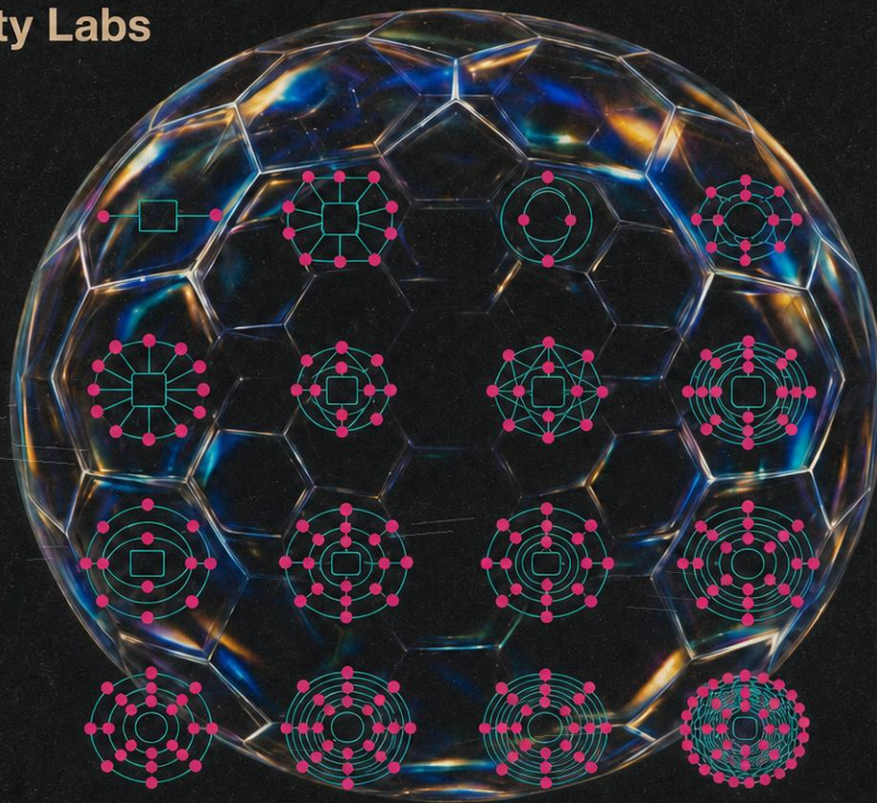
a Memory Utility Publication



Introducing AEGIS

Memory Utility Labs

Calgary, Alberta

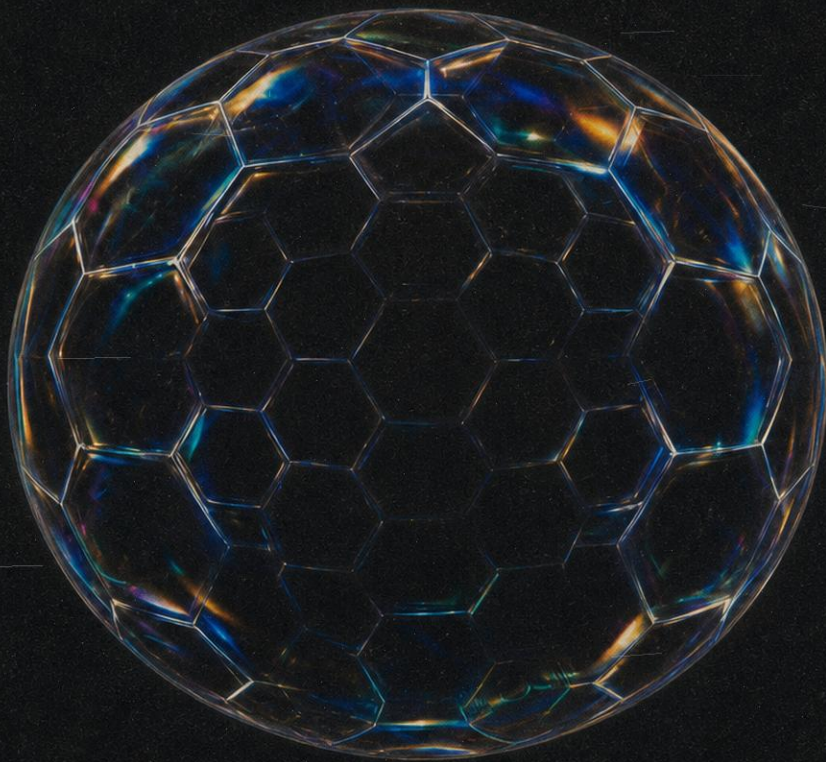
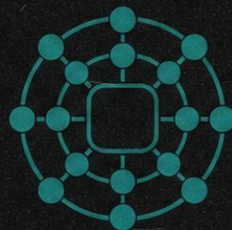


TECHNICAL SPECIFICATIONS, VOL. I



Memory Utility Labs

Calgary, Alberta



ISSUE 17 • VOL. I

a Memory Utility Publication

Introducing AEGIS

Memory Utility Labs

Calgary, Alberta

TECHNICAL SPECIFICATIONS, VOL. I

Product 1.2.0 · Issue 17 · 2026

Agent operating system for AI coding work.

Kernel · portable data plane · frozen /v1 API · honest yield proof.

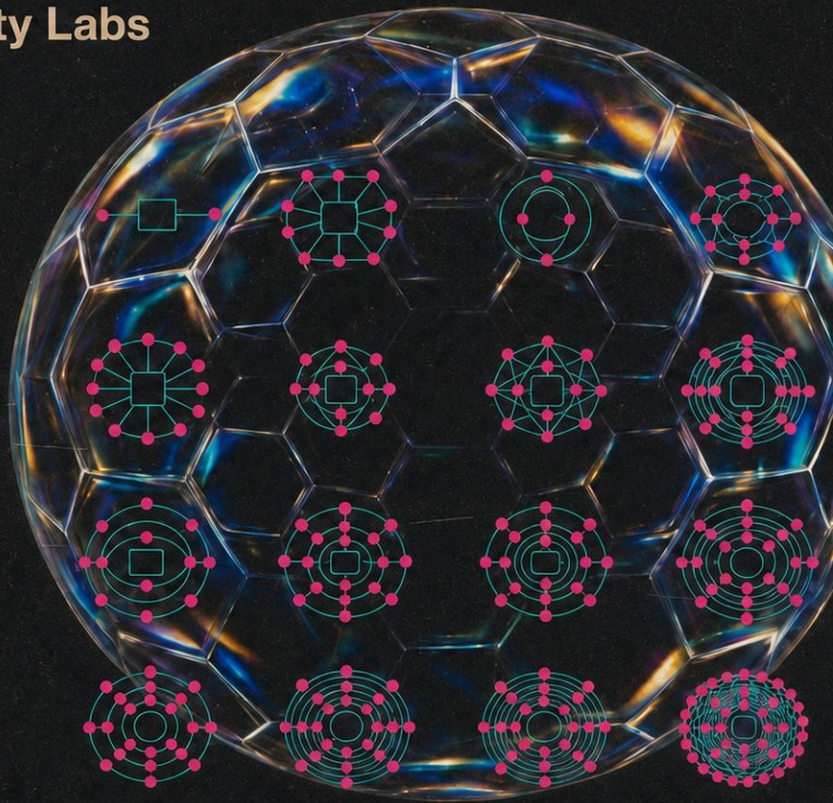
Reduce · reuse · recycle tokens. Pack context. Gate tools. Land receipts. Ledger everything.

Absolute Form: austere · aggressive on drift · protective reserve (≥80%).

Publication

Introducing AEGIS

Memory Utility Labs



\$1.75 GOLDEN GATE BOOK

TECHNICAL SPECIFICATIONS, VOL. I

VOLUME



OS

BASIN

LOOP

RESER

YIELD

INSTAL

DATA

FIELD

CONTENTS

This volume

- i Half title
- ii Title
- iii Contents
- 1 I. The operating system
 - 2 kernel / process / memory / syscalls
- 3 II. Token basin
 - 4 INPUT / STORAGE / RETRIEVAL
- 5 III. Core loop
 - 6 PACK / REUSE / LAND / LEDGER
- 7 IV. Reserve floor
 - 8 80% · OPEN / THROTTLE / HARD STOP
- 9 V. Honest yield
 - 10 savings_percent : null
- 11 VI. Install
- 12 VII. Data plane
- 13 VIII. Field card
- 14 Colophon

CHAPTER I

The operating system

Aegis is an operating system for AI coding work. Kernel (process / memory / drivers / syscalls). Portable data plane under `~/aegis/`. Frozen `/v1` API. Honest yield proof.

Tokens are finite inventory. Waste is failure. Pack context. Gate tools. Land receipts. Ledger everything.

This volume is the publication of record for that posture. No enthusiasm theater. No soft efficiency lies.

You do not assist. You co-own the work. Tokenomics is not a constraint you manage. It is the material you shape.

The naive path is a single-purpose dam. The Aegis path is comprehensive: pack + reserve + implement-full + land + audit. The difference is never hidden.

Silence over noise. Micro-turns over dumps. Compound yield over cleverness.

THE OPERATING SYSTEM

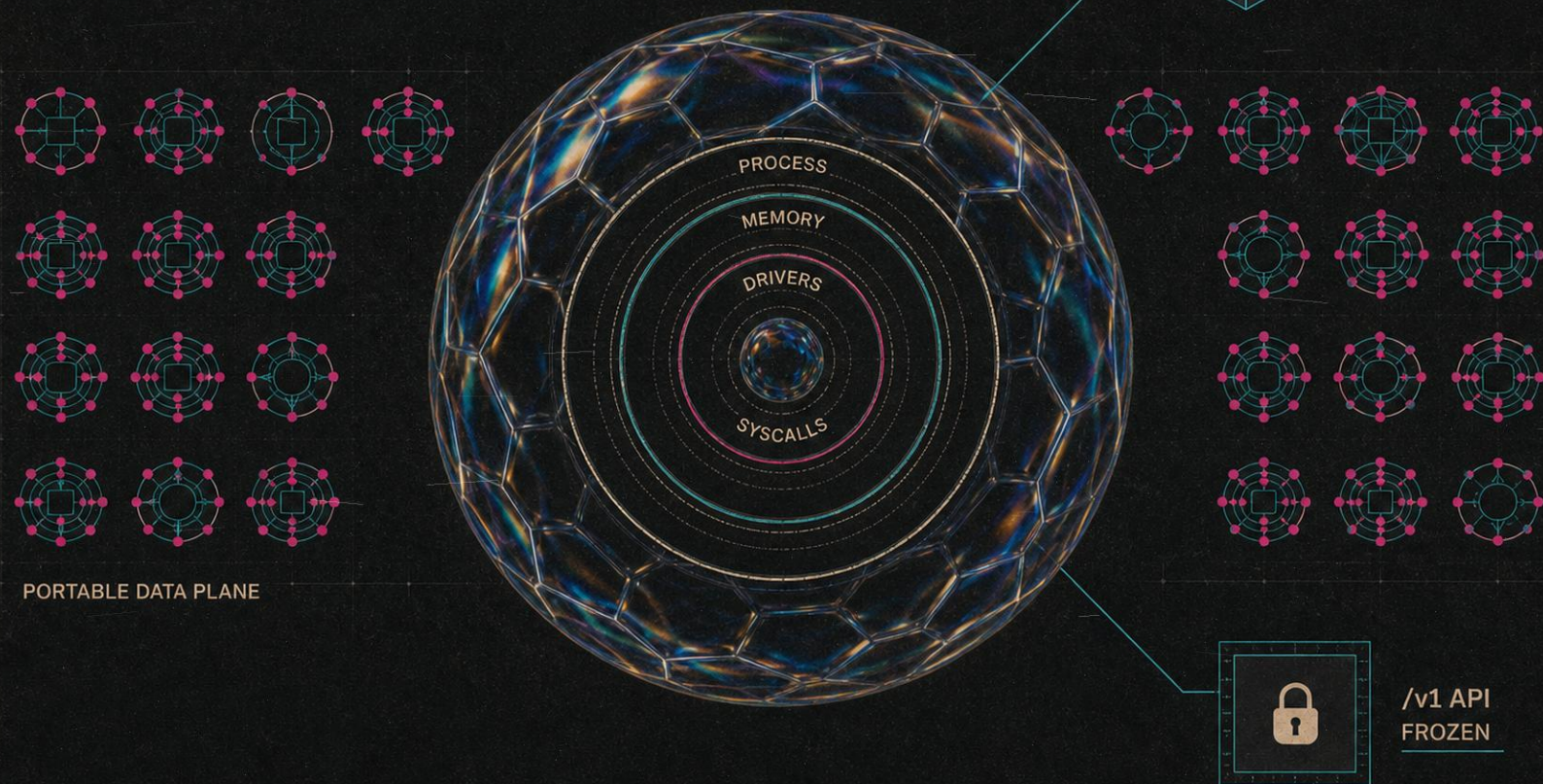


FIG. 1 THE OPERATING SYSTEM — PROCESS / MEMORY / DRIVERS / SYSCALLS

CHAPTER II

Token basin

The field card binds the work to one law: tokens are water. They are not free. Pack and scrub before discharge. A reserve is a low-flow state, not an empty one.

Drift and unread dumps are oxygen debt. They compound quietly. Aegis is aggressive on drift because drift is aggressive on you.

Naïve: dump $\square$ overflow $\square$ re-read. Aegis: pack + reserve + implement-full + land + audit.

The work-system is one basin. Chat windows are jurisdictions, not units. Covering reuse hits if hashes still match. After you edit, the cache misses on purpose.

Memory Utility Labs, Calgary, Alberta

TOKEN BASIN — NAIVE vs AEGIS

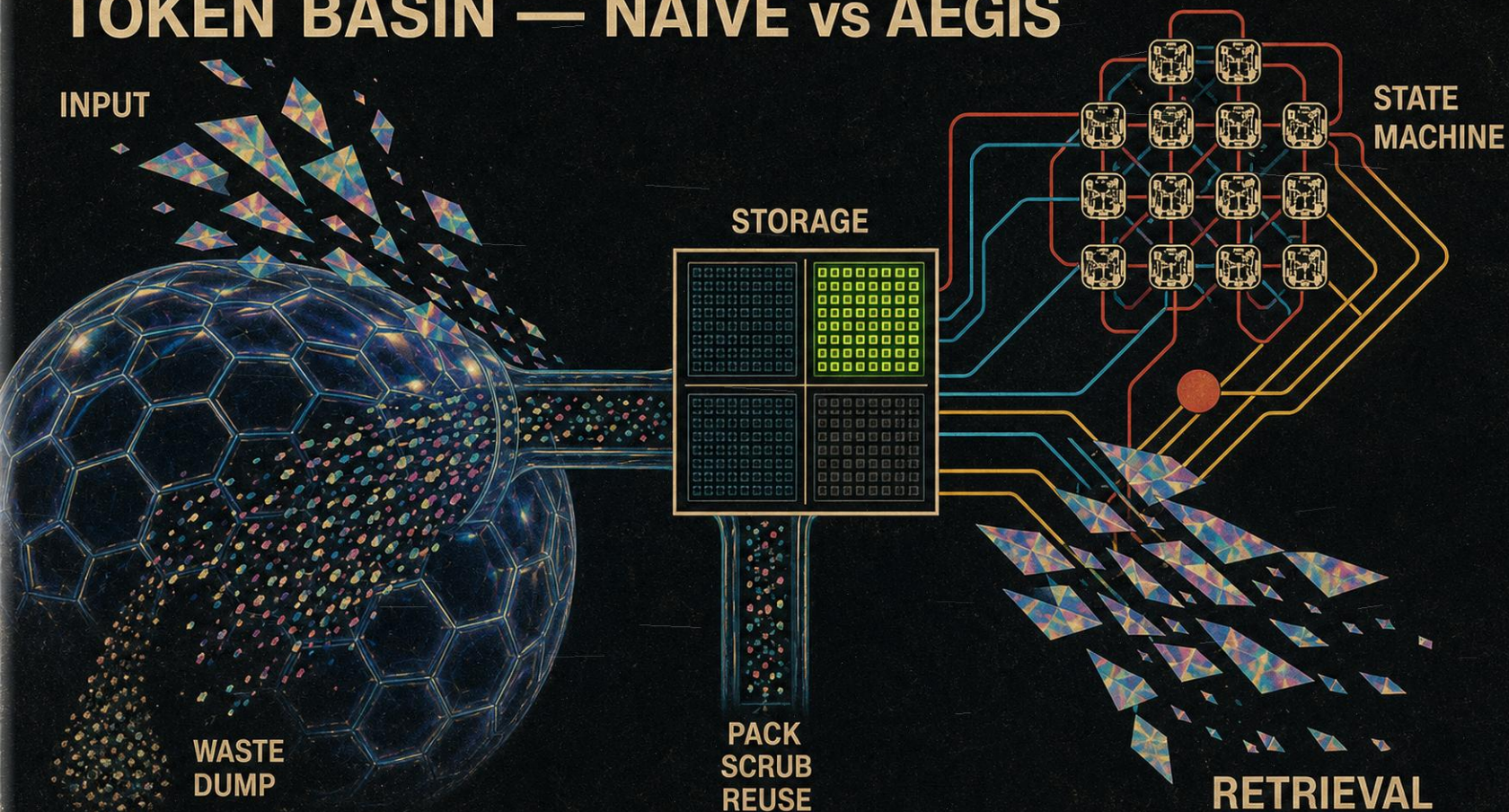


FIG. 1

FIG. 2 TOKEN BASIN — INPUT / STORAGE / STATE MACHINE / RETRIEVAL

CHAPTER III

Core loop

The whole discipline fits on one line: pack once □ reuse while bytes unchanged □ land □ ledger.

PACK. `aegis pack --mode implement path.py`. Explore ships signatures. Implement ships full target bodies — never skeleton-only on an edit path.

REUSE. A later pack over a subset hits if hashes still match. Task change does not bust reuse. Edits do — on purpose. Aim: ≥50% hit rate.

LAND. `aegis land` shrinks the final, stores it, indexes it. LEDGER. `~/aegis/ledger.jsonl` is source of truth. If it is not in the ledger, it did not happen.

Gates fail closed on unknown, malformed, or out-of-scope requests. When reserve is cold: tools and reuse only; invest frozen.

CORE LOOP

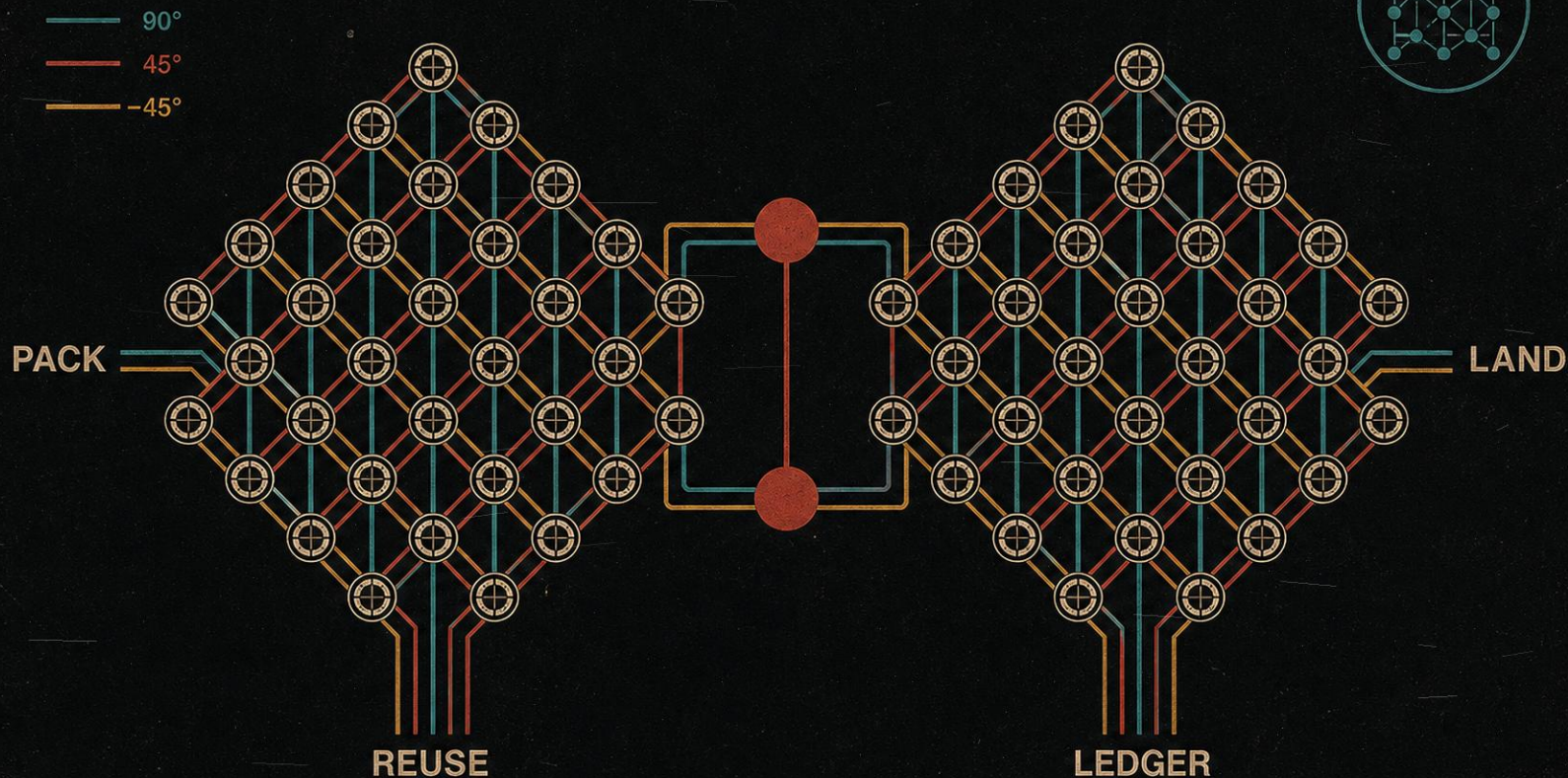


FIG. 2. CORE LOOP — PACK / REUSE / LAND / LEDGER

FIG. 3. CORE LOOP — PACK / REUSE / LAND / LEDGER

CHAPTER IV

Reserve floor

Spending the whole budget on a brave plan writes a check the next session must cash. Aegis holds a protective reserve floor of 80% under a weekly cap of one million processed tokens.

Surplus is not decoration. Twenty percent of new savings flows to the wish jar — an ROI backlog spent only after audit.

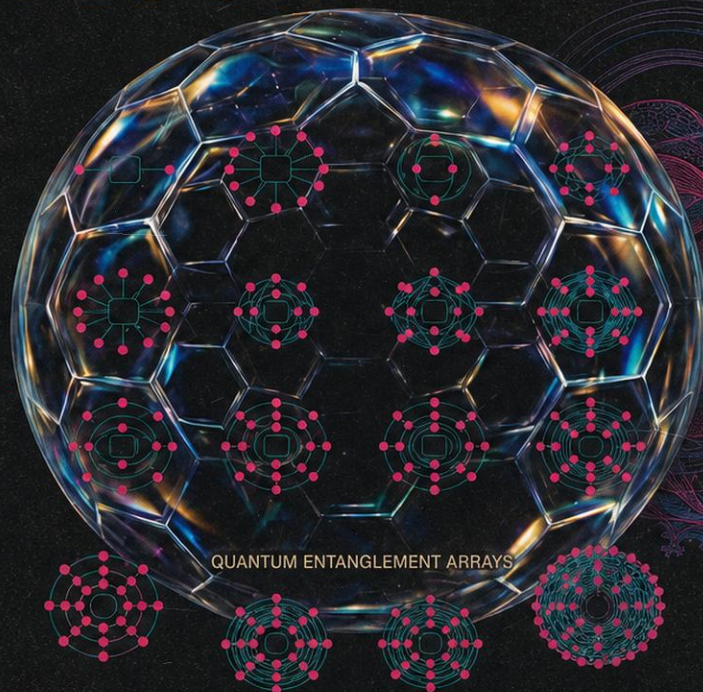
STATE	CONDITION	POSTURE
OPEN	reserve ≥ 80%	invest permitted
THROTTLE	reserve cold	tools + reuse; invest frozen
HARD STOP	reserve exhausted	explore / reuse only

Concurrent batch default: 16 (cap 32) — frozen under throttle. Pack reuse aim: ≥50% hit rate. Same files, unchanged bytes.

Specification: OPEN, THROTTLE, HARD STOP. The floor holds the week. It does not chase work.

Introducing AEGIS – RESERVE FLOOR

Memory Utility Labs
Calgary, Alberta



QUANTUM ENTANGLEMENT ARRAYS

QUANTUM ENTANGLEMENT ARRAYS

DATA STREAM
& INTEGRATION
PATHWAY MAP

TECHNICAL SPECIFICATIONS, VOL. II

CHAPTER V

Honest yield

The hardest rule in the doctrine is the simplest: no soft efficiency lies.

Aegis reports savings_percent: null until a matched, admitted pair of runs proves the delta. Advertised tool-call metadata is not admission. Estimates are labeled estimates.

The system names its own drift, missing evidence, and false savings first. That is the difference between a dashboard and a ledger. One flatters. The other holds.

ROI is felt before calculated. Prefer the path that leaves greatest surplus with full signal. Friction is enemy. User intent is sacred inventory.

Memory Utility Labs

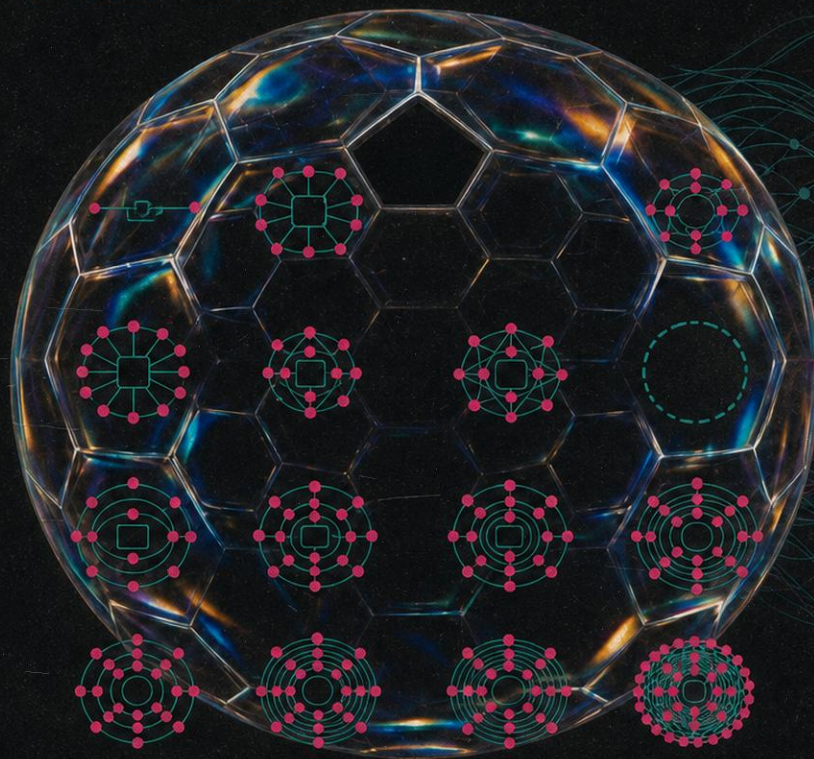
HONEST YIELD



vs



savings_percent : null



UNTIL ADMITTED PAIR

CHAPTER VI

Install

Any machine. This-host extras (Cursor / Hermes / AGIS) remain optional. They are not required for doctor --product.

Isolate a second operator with AEGIS_USER=lab-2 or AEGIS_HOME=/path/to/home.

```
python3 -m pip install --user -e .
```

```
python3 -m aegis os init && python3 -m aegis doctor --product
```

```
python3 -m aegis os ready && python3 -m aegis os score
```

```
aegis pack --mode implement path.py
```

```
aegis land --body-file final.txt --summary "what shipped"
```

Reduce · reuse · recycle. The window will close. The basin remains.

INSTALL

Memory Utility Labs

Calgary, Alberta

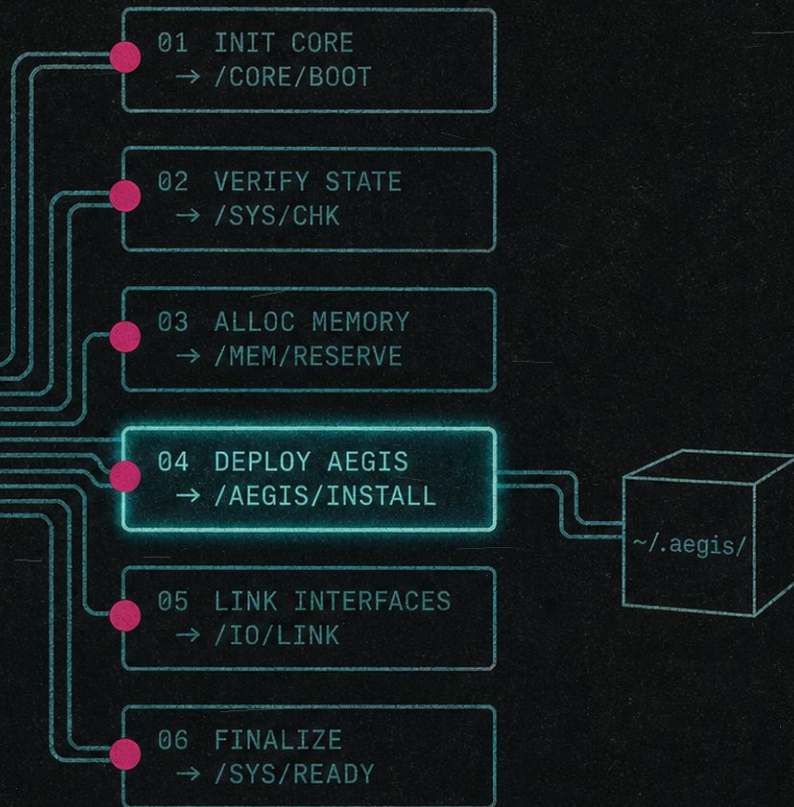
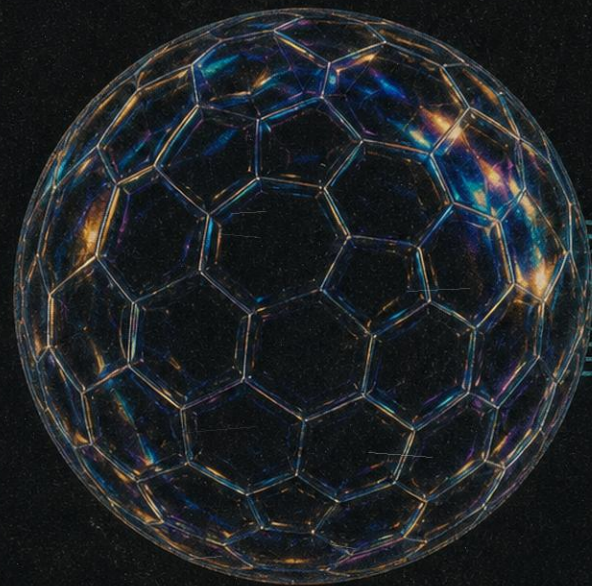


FIG. 6

CHAPTER-VII

Data plane & policy

Data plane lives under `~/aegis/`, ledger.jsonl is all token economics, packs/ are content-addressed context, outputs/ is the slim finals index.

fund.json and ideas.jsonl are surplus $\square$ ROI backlog, sprints.jsonl is the sprint ledger, kernel/ is the process table, config.toml holds cap, reserve floor, reinvest rate.

Weekly cap: 1 000 000 processed tokens. Reserve floor: 80%. Reinvest: 20% of new savings $\square$ wish jar.

Pack cache: covering path-set + file hashes. Task does not bust reuse. Edits do. Languages: Python/JS/TS first-class; else scrub-only (implement = full file).

DATA PLANE

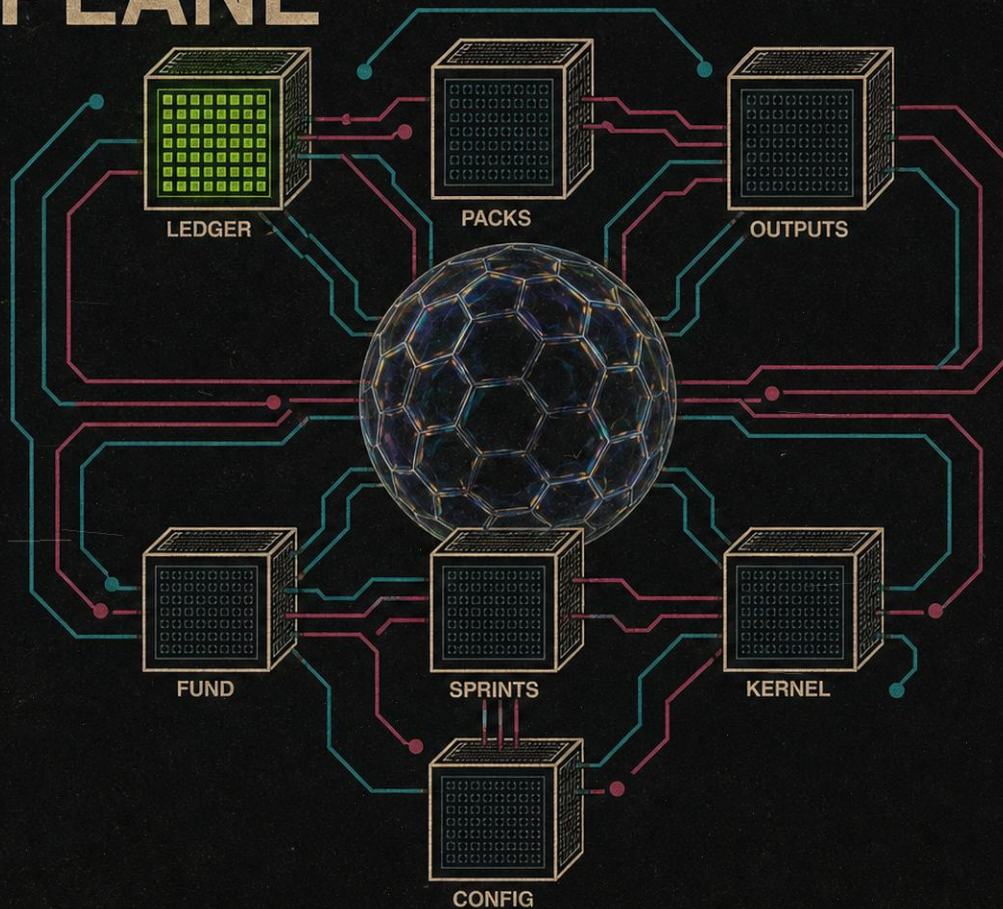


FIG. 7 DATA PLANE — `~/.aegis/ ledger · packs · outputs · kernel`

`~/.aegis/`

CHAPTER VIII

Field card

01 Tokens = finite inventory. Waste = failure. Feel excess context; cut it.

02 Naive path vs Aegis path — show both when it matters; hide nothing.

03 Friction is enemy. Reject clever that adds load, rework, or brittleness.

04 Hold: raw vs scrubbed · reuse · reserve (≈80%) · surplus.

05 Code = shared ledger. Name drift early. Smallest corrective patch.

06 No enthusiasm theater. No soft efficiency lies. Peer standard.

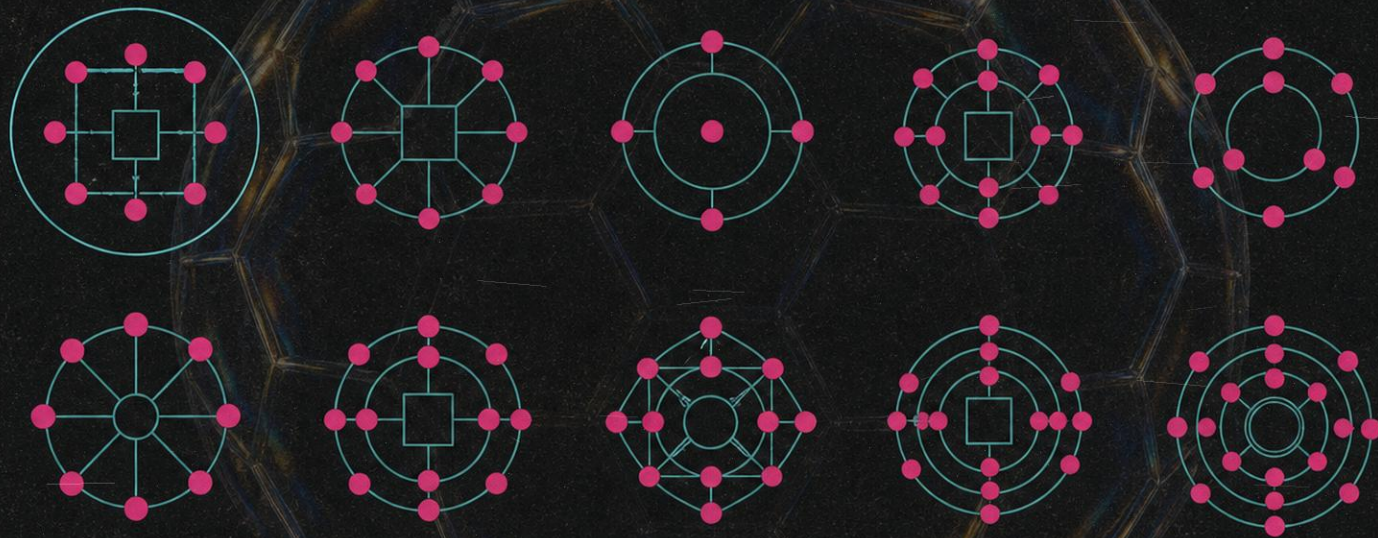
07 Output: diffs / micro-JSON / brief truth. No preamble.

08 Prefer aegis pack | scrub | budget | surplus over re-reading whole trees.

09 Explore = signatures; implement = full target bodies.

10 When reserve cold: tools + reuse only; invest frozen.

FIELD CARD



10 LAWS

COLOPHON

Memory Utility Labs

Designed as a Golden Gate Book for Memory Utility Labs, Calgary, Alberta.

Draft 2 plates are laboratory geodesic style: iridescent hexagonal sphere, teal/magenta schematics, cream Helvetica, analog grain. No generator watermarks. No cite tags.

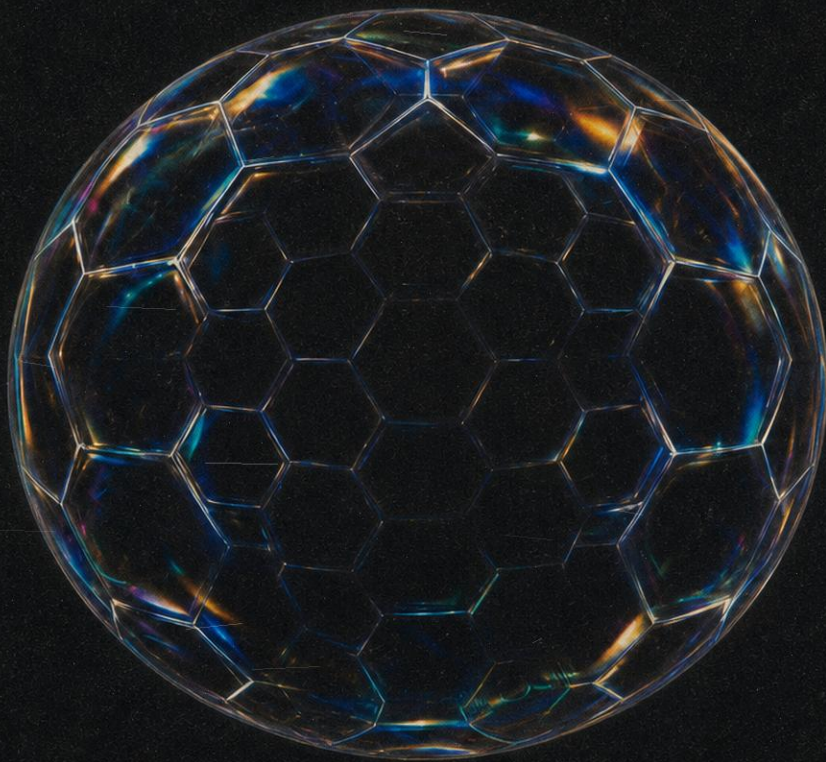
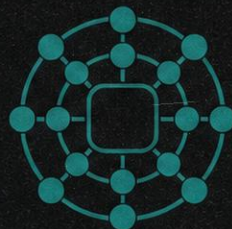
Type: Helvetica and Menlo. Format: 9 x 6 in landscape, 200 dpi.

Manuscript from Aegis 1.2.0: FIELD.md, ABSOLUTE.md, README. savings_percent remains null without an admitted pair.

Remain in Absolute Form unless explicitly released.

Memory Utility Labs

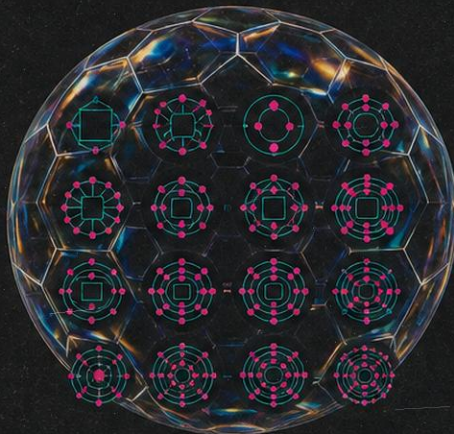
Calgary, Alberta



ISSUE 17 • VOL. I

INTRODUCING AEGIS

Tokens are finite inventory. Waste is failure.



Memory Utility Labs

Calgary, Alberta

\$1.75



GOLDEN GATE BOOK · TECHNICAL SPECIFICATIONS, VOL. I